2.6 Derivatives of e? and ln x

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

What is the slope of e? at every point, and of ln x?

Targets

- I can state the derivatives of e? and ln x.
- I can differentiate sums that use both rules.
- I can find slopes, tangent lines and normal lines.

Concepts

- **The curve e?** (d/dx e? = e?): Its slope at a point equals its height there. It never has a horizontal tangent.
- **The curve ln x** (d/dx ln x = 1/x): Its slope at x is the reciprocal of x. The slopes shrink as x grows: it flattens out.
- **Tangent and normal** (normal slope = $-1/m$): Tangent: $y - y_0 = m(x - x_0)$. Normal: slope is the negative reciprocal, $-1/m$.

What the class predicted, and what is true

- The curve $y = e^x$ climbs faster and faster. As x increases, its slopes will... **keep growing.** The curve gets steeper, so its slopes keep growing.
- The curve $y = \ln x$ rises but flattens out. As x increases, its slopes will... **shrink toward 0.** A flattening curve has slopes that shrink toward 0 (they are $1/x$).

Huddle notes (answers filled in)

Definitions

- **e?:** Its slope at any point equals its **height**: d/dx e? = **e?**.
- **ln x:** Its slope at x is the **reciprocal** of x: d/dx ln x = **1/x**.
- **Horizontal tangent:** The slope is **zero**. The curve e? never has one.

Worked example

Find the slope of $f(x) = \ln x + 3x^2$ at x = 1.

- f?(x) = 1/x + **6x**
- f?(1) = 1 + **6**
- So the slope is **7**.

Label the curve e?

1. The **point** where we measure.
2. The **tangent line**: its slope equals the height.
3. The curve is **y = e?**.

One worked sample of each skill

Differentiate

Level 1. Differentiate.
Answer: e^x $\dfrac{d}{dx}e^x = e^x$.

Level 2. Differentiate.
Answer: $\frac{2}{x}$ $2 \cdot \dfrac{1}{x} = \dfrac{2}{x}$.

Level 3. Differentiate term by term.
Answer: $10x + 2e^x - \frac{5}{x}$ $10x + 2e^x - \dfrac{5}{x}$.

Level 4. Differentiate. Rewrite the root as a power if that helps.
Answer: $\frac{6}{2\sqrt{x}} - \frac{6}{x} + e^x$ $\dfrac{6}{2\sqrt{x}} - \dfrac{6}{x} + e^x$.

Slope at a point

Level 1. Find the slope of $y = e^x$ at $x = 0$.
Answer: 1 The slope equals the height, $e^0 = 1$.

Level 2. Let $f(x) = 6e^x + 5x$. Find $f'(0)$.
Answer: 11 $f'(x) = 6e^x + 5$, so $f'(0) = 6 + 5 = 11$.

Level 3. Let $f(x) = 6\ln x + x^2$. Find $f'(4)$.
Answer: 9.5 $f'(x) = \dfrac{6}{x} + 2x$, so $f'(4) = 9.5$.

Level 4. Let $f(x) = 6\ln x - \dfrac{6}{x}$. Find $f'(2)$.
Answer: 4.5 $f'(x) = \dfrac{6}{x} + \dfrac{6}{x^2}$, so $f'(2) = 4.5$.

Tangent and normal lines

Level 1. Find the tangent line to $y = \ln x$ at $x = 1$.
Answer: $x - 1$ Point $(1, 0)$, slope 1, so $y = x - 1$.

Level 2. Find the tangent line to $y = 2e^x$ at $x = 0$.
Answer: $2 + 2x$ Point $(0, 2)$, slope 2, so $y = 2 + 2x$.

Level 3. Find the tangent line to $y = e^x - 2x$ at $x = 0$.
Answer: $1 + (-1)\,x$ Point $(0, 1)$, slope -1: $y = 1 + (-1)x$.

Level 4. Find the normal line to $y = \ln x + 4x^2$ at $x = 1$.
Answer: $4 - \frac{x-1}{9}$ Tangent slope 9, so the normal slope is $-\frac{1}{9}$: $y = 4 - \dfrac{x-1}{9}$.

Limits that are derivatives

Level 1. Evaluate $\lim_{h\to 0} \dfrac{\ln(2+h) - \ln 2}{h}$.

Answer: 0.5 It is $f'(2)$ for $f = \ln x$: $\dfrac{1}{2}$.

Level 2. Evaluate $\lim_{h\to 0} \dfrac{e^h - 1}{h}$.

Answer: 1 It is $f'(0)$ for $f = e^x$: $e^0 = 1$.

Level 3. Evaluate $\lim_{x\to 5} \dfrac{\ln x - \ln 5}{x - 5}$.

Answer: 0.2 It is $f'(5)$ for $f = \ln x$: $\dfrac{1}{5}$.

Level 4. Evaluate $\lim_{x\to 0} \dfrac{6(e^x - 1)}{x}$.

Answer: 6 $f(x) = 6e^x$, $f(0) = 6$, so the limit is $f'(0) = 6$.

Horizontal tangents

Level 1. Let $f(x) = 2\ln x - x$ for $x > 0$. At which x is the tangent horizontal?

Answer: 2 $f'(x) = \dfrac{2}{x} - 1 = 0$ gives $x = 2$.

Level 2. Let $f(x) = 10\ln x - 2x$ for $x > 0$. At which x is the tangent horizontal?

Answer: 5 $\dfrac{10}{x} - 2 = 0$ gives $x = 5$.

Level 3. Let $f(x) = 6\ln x - 3x + 7$ for $x > 0$. At which x is the tangent horizontal?

Answer: 2 $f'(x) = \dfrac{6}{x} - 3 = 0$ gives $x = 2$.

Level 4. Let $f(x) = \ln x + \dfrac{7}{x}$ for $x > 0$. At which x is the tangent horizontal?

Answer: 7 $f'(x) = \dfrac{1}{x} - \dfrac{7}{x^2} = 0$ gives $x = 7$.